

Week 1 (DSA)

Exercise 7: Financial Forecasting

```
package com.cognizant.forecasting;

public class FinancialForecast {

    // Recursive method to calculate future value
    public static double calculateFutureValue(double currentValue, double
growthRate, int years) {

        // Base case
        if (years == 0) {
            return currentValue;
        }

        // Recursive call
        return calculateFutureValue(currentValue, growthRate, years - 1) * (1
+ growthRate);
    }

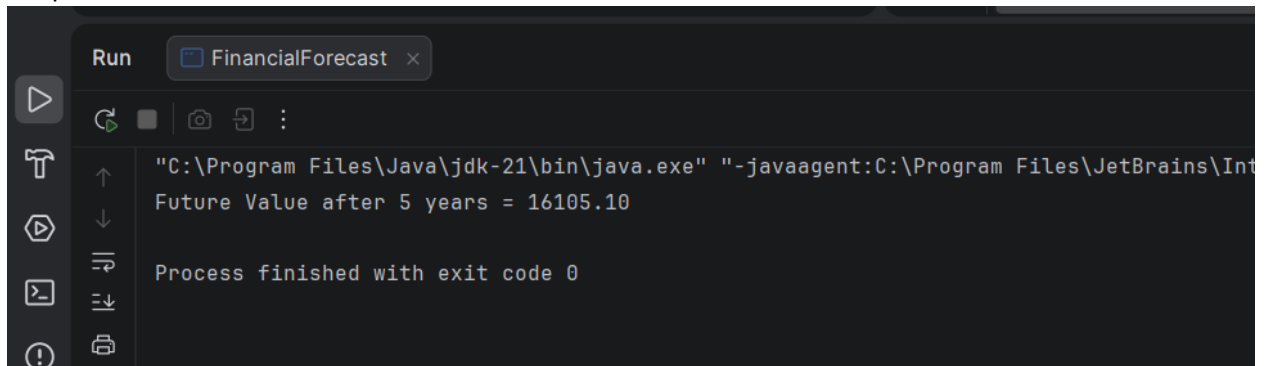
    public static void main(String[] args) {

        double currentValue = 10000;    // Initial Investment
        double growthRate = 0.10;       // 10%
        int years = 5;

        double futureValue = calculateFutureValue(currentValue, growthRate,
years);

        System.out.printf("Future Value after %d years = %.2f%n", years,
futureValue);
    }
}
```

Output



The screenshot shows the 'Run' console of an IDE. At the top, there's a tab labeled 'FinancialForecast' with a close button. Below the tab, there's a toolbar with icons for running, debugging, and other actions. The main area of the console displays the command executed: `"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\Int"`. The output of the program is `Future Value after 5 years = 16105.10`. At the bottom, it states `Process finished with exit code 0`. On the left side of the console, there's a vertical toolbar with icons for running, debugging, and other actions.